

Week 5 Homework (April 2)

Due date: April 7 in class

1. Section 9.3: 4, 7, 10, 11, 15, 16.
2. Let $\mathrm{GL}(n, \mathbb{R}) \subset \mathbb{R}^{n^2}$ be the set of $n \times n$ invertible matrices (by identifying $A = (x_{ij})$ with the vector $(x_{11}, \dots, x_{1n}, x_{21}, \dots, x_{nn})^T$).
 - (a) Using the map $\det : \mathbb{R}^{n^2} \rightarrow \mathbb{R}$ to prove $\mathrm{GL}(n, \mathbb{R})$ is open. As a consequence, the small perturbation of an invertible matrix is also invertible.
 - (b) Show the map $\varphi : \mathrm{GL}(n, \mathbb{R}) \rightarrow \mathrm{GL}(n, \mathbb{R})$ sending A to A^{-1} is continuous.
 - (c) Show that φ is differentiable, and find $d\varphi|_A$.